PIVoT: Part-Wise Identification of Volume-Aware Density via Multi-Topology Reorientation

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***Abstract*—Reproducing articulated objects faithfully in simulation is essential for learning robotic manipulation, and an object's physical behavior is determined by each part's volume and density, so both must be identified at the part level. Existing vision-based methods, however, estimate volume only at the whole-object level and infer density from appearance alone, while interaction-based identification has been applied only to rigid bodies; the design of measurement poses and paths for identifying the part-wise parameters of an articulated object has not been addressed. We present PIVoT, a closed-loop pipeline that identifies the per-part density of articulated objects through active robot interaction and produces sim-ready assets faithful in appearance, geometry, and physics. Based on information gain, PIVoT suggests to the user joint configurations that minimize the null space of the density estimate; at each configuration, the robot collects wrench data along collision-free paths through three measurement poses under different gravity directions and updates the per-part densities until their uncertainty converges. User intervention beyond reconstruction is limited to adjusting the object to the suggested joint configuration, and the per-part volumes on which the identification rests are corrected by multi-view-consistency-based mesh refinement and per-part scaling. With a single RGB-D camera and a 6-DoF arm, we validate PIVoT in simulation on three real articulated objects, including two custom objects with per-part ground-truth density; validation on a fourth object (laptop) and on the physical robot is in preparation.**

***Index Terms*—Articulated object, Physical property identification, Real2Sim**

I. Introduction

Physics simulation has become a driving force in robotic manipulation, letting policies be trained and evaluated before real-world deployment. The real world is full of portable articulated objects—laptops, foldable phones, desk lamp stands—whose friction-holding joints keep their configuration unless a force is deliberately applied. For simulation with such objects to transfer, sim-ready assets must be dynamically as well as visually and kinematically faithful; hand-building them does not scale [1], motivating real-to-sim pipelines that construct them from real observations.

Recent 3D Gaussian Splatting (3DGS) pipelines recover an articulated object's appearance, part-level meshes, and joint structure up to a URDF [2]. Dynamically faithful simulation additionally requires each part's mass, center of mass, and rotational inertia, all of which follow from

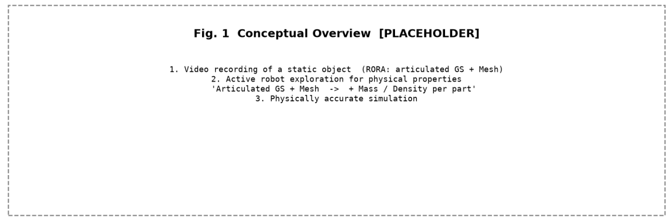

Fig. 1. PIVoT overview. (1) Video of a static object yields an articulated reconstruction; (2) robot reorientation under different gravity directions adds per-part mass and density; (3) the asset reproduces the object's behavior in simulation.

per-part geometry and density. Reconstruction supplies the geometry; density is precisely what a camera cannot observe, since identical-looking parts can hide entirely different mass distributions. Existing pipelines therefore guess density—from appearance, material priors, or a generative model [3]–[5]—while methods that measure mass through real interaction keep the object rigid [1], [6]. To our knowledge, no prior work has identified, from wrist force/torque measurements, the density of the links themselves for an articulated object in which two or more links carry independently unknown mass: interaction-based attempts on articulated objects have estimated link masses from vision rather than force sensing, and force-based identification has been demonstrated only on mechanisms with a single movable link [7].

The gap persists not because the rigid-body methods are weak, but because the measurement they rely on saturates. A quasi-static wrench constrains only the zeroth and first mass moments: for an object with P parts a single joint configuration determines at most $1 + \min(3, P)$ independent density combinations—at most four—however the wrist reorients the object. Changing the object's own joint configuration is therefore the only mechanism that brings the remaining combinations into view, and the same analysis yields a lower bound on the number of configurations required, computable from geometry alone, before any wrench is measured.

Turning this into a working system requires two ingredients the rigid-body literature never needed. The first is choosing *which* configurations to measure: only a handful are affordable, so each should maximize expected information gain about the per-part densities. The second is physically *reaching* the measurement poses: each configuration is measured under three gravity directions, and the robot must carry the grasped object between poses without collision and without overloading its joints, across topologies whose geometry—hence collision constraints—differ. Moreover, the joint configuration is read by a camera rather than commanded, so the design variable itself carries error; a total-least-squares formulation absorbs it, and conditioning—not the rank condition—sets the practical limit.

We therefore propose PIVoT (Fig. 1), a closed-loop pipeline that identifies the per-part density of portable articulated objects through active robot interaction, turning the part-level reconstruction of [2] into a sim-ready asset faithful in appearance, geometry, and physics. The reconstruction-stage intervention is inherited from [2]; the additional intervention for physics estimation is one adjustment of the object to the suggested joint configuration. Our contributions are:

- **Information-theoretic part-wise density identification:** a lower bound on the number of joint configurations as a function of the part count, selection by expected information gain, and an uncertainty-aware per-part density update.
- **Multi-topology measurement path design:** three measurement poses under different gravity directions per configuration, linked by RRT-planned collision-free paths that account for object geometry, self-collision, and obstacles across topologies.
- **Closed-loop sim-ready asset pipeline:** mesh refinement and per-part scaling reduce volume error, and the components above integrate into a pipeline limiting additional user intervention to one joint-configuration adjustment, validated in simulation on three articulated objects, two with per-part ground-truth density; validation on a fourth object (laptop) and on the physical robot is in preparation.

II. Related Work

A. Sim-Ready Reconstruction of Articulated Objects

Articulated-object reconstruction recovers appearance, part decomposition, and kinematics from interaction or images [8]. RORA, our starting point, exports part-bound Gaussians, part meshes, and a URDF from a single-configuration video through human-in-the-loop joint proposals [2]. ArtVIP's assets are billed as physically faithful but hand-authored, not identified from measurements of the simulated instance [9]; JODA infers joint dynamics—friction, damping, detents—with a VLM, leaving link masses unaddressed [10]. The lineage delivers visual and geometric fidelity but no measured physics; we take such a reconstruction as input and add physics by measurement.

B. Vision-Based Physical Property Estimation

Vision-based methods estimate properties without contact. For *volume*, PUGS integrates a Gaussian-splatting reconstruction's volume to predict object mass [3], and the PhysX series uses its generated meshes' volume [5]. Volume thus enters at whole-object granularity or via generated geometry; an individual part—thin, hollow, or occluded—retains a large error.

For *density*, the generative series looks up a material database given a predicted material class, while PhysGS and part-level SiPhy infer it from appearance and language-derived priors [4], [11]. These estimates are grounded only in appearance priors: real measurements enter as evaluation ground truth—never as an input. Appearance does not determine density; we identify it by measurement instead.

C. Interaction-Based Physical Property Estimation

Interaction-based methods measure, but on single rigid bodies: quasi-static wrenches identify masses of homogeneous material regions inside one body—the "parts" of [6] are such regions, not articulated links—inertial parameters are identified during pick-and-place [12], and Scalable Real2Sim generates assets with physically feasible inertial parameters from robot sensing [1].

A second thread reorients a grasped rigid body, reading the static wrench: one gravity direction fixes mass, two non-parallel add the center of mass, a non-coplanar third separates sensor bias; the inertia tensor produces no quasi-static wrench and classically requires dynamic trajectories [13], [14] (cf. Sec. III-C).

The closest prior work is Kumar et al. [15], which estimates per-link mass of two- and three-link objects on a real UR10; it differs in sensing with vision (an RGB-D camera and per-link markers) rather than force–torque, in pushing the object non-prehensilely rather than grasping and reorienting it, and in classifying among five discrete mass-distribution hypotheses rather than regressing continuous values. RigPI identifies parameters via VLM-seeded differentiable simulation with force–torque sensing [7], but its real-robot validation objects each have only one moving link; its multi-link results remain in simulation. Interactive perception acts on articulated objects but estimates the kinematic model—joint axes and types—not link masses or densities [16]. To our knowledge, no prior work identifies, from wrist force–torque measurements, the densities of two or more independently unknown links within one mechanism.

Our design machinery is classical: base-parameter analysis delimits which inertial combinations any regressor can resolve [17]; Gautier–Khalil and Swevers et al. condition the regressor by optimizing identification trajectories [18], [19]; ASID brings the same logic to learned exploration policies maximizing Fisher information [20]. All design an actuated robot's trajectory, its configuration read by encoders; PIVoT instead designs, in a closed loop, a passive object's joint configuration and the gravity directions under which it is reoriented.

Fig. 2 Overall Pipeline [PLACEHOLDER]

Top (offline, grey): (a) Articulated reconstruction [RORA, human-in-the-loop]
(b) Volume estimation (mesh refinement + per-part scaling)
Bottom (purple dashed loop): (c) Joint configuration suggestion [user]
(d) Measurement pose & path design
(e) Active robot sensing (f) Density update
(f)->(c) not converged (f)->(g) converged
Right (teal): (g) Sim-ready asset

Fig. 2. Pipeline. From the asset of [2], the offline stage estimates part volumes; the closed loop suggests joint configurations, reorients the object to three gravity directions, and updates part densities until convergence.

III. Method

A. Problem Formulation and Overview

RORA [2] reconstructs the object from a video of one static configuration: P rigid parts, each with a mesh and 3D Gaussians, in a metric-scale URDF. Part i has volume V_i and unknown uniform density ρ_i; mass, center of mass, and inertia follow analytically, so $\rho = (\rho_1, \ldots, \rho_P)^\top$ alone separates a visually faithful asset from a dynamically faithful one.

A gripper rigidly grasps the object behind a wrist force/torque sensor (frame $\mathcal{S}$); the configuration $\theta \in \Theta \subset \mathbb{R}^J$ is friction-held, operator-set, and camera-read—known only up to measurement error. The wrist is quasi-statically reoriented through gravity directions $\hat{g}$, reading the wrench $y = (f, \tau)$ after tare.

The wrench is linear in ρ: with

$$v \triangleq (V_1, \ldots, V_P)^\top, \;\; C(\theta) \triangleq \begin{bmatrix} V_1 c_1(\theta) & \cdots & V_P c_P(\theta) \end{bmatrix} \in \mathbb{R}^{3\times P}, \tag{1}$$

where $c_i(\theta)$ is the centroid of part i in $\mathcal{S}$,

$$y \;=\; A(\theta, \hat{g})\,\rho + \varepsilon, \qquad A(\theta, \hat{g}) = g \begin{bmatrix} \hat{g}\, v^\top \\ -[\hat{g}]_\times C(\theta) \end{bmatrix} \in \mathbb{R}^{6\times P}. \tag{2}$$

A needs volumes and centroids only—no density; noise is $R = \mathrm{blkdiag}(\sigma_f^2 I_3, \sigma_\tau^2 I_3)$; one weighing of the assembled object gives $v^\top \rho$, seeding the prior.

The estimation vector must count every mass the wrench feels: small, occluded hardware—above all the friction hinges holding θ, 41 g in 7.45 cm^3 on our custom objects—enters ρ as its own component. Omitted, the hinge mass biases by 79.1% while the reported half-width stays 0.83%—confidently wrong, unrepaired by more rounds.

PIVoT closes a loop around this model (Fig. 2): an offline stage (Sec. III-B) produces the volumes and centroids behind every regressor; each round selects the most informative joint configuration—set by hand by the operator, the only intervention added beyond those inherited from reconstruction—then measures along designed poses and paths and updates the posterior over ρ until a target relative uncertainty is met.

B. Part-Wise Volume Estimation

This stage turns the captured views and RORA's meshes into the V_i and $c_i(\theta)$ of (1): mesh refinement enforces multi-view consistency [pending: refinement implementation from collaborator], then each part is rescaled along its principal (PCA) axes against the anisotropic scale error that survives [pending: refinement implementation from collaborator]. Volumes and centroids are integrated from the result (non-watertight meshes via their exactly integrable convex-decomposition union); CAD solids and direct measurement serve as ground truth, never pipeline inputs.

Remark 1 (*Volume invariance*): In (1)–(2) the measurement depends on ρ only through the products $V_i \rho_i$: for fixed centroids, replacing each V_i by any $V_i' > 0$ and ρ_i by $(V_i/V_i')\rho_i$ leaves $A(\theta, \hat{g})\rho$ unchanged. The estimated part masses $\hat{m}_i = V_i \hat{\rho}_i$ are therefore invariant to volume error, which enters only the derived density $\hat{\rho}_i = \hat{m}_i / V_i$.

The invariance is exactly that narrow—its limits are what "volume-aware" means: density inherits volume error one-to-one ($\hat{\rho}_i = \hat{m}_i / V_i$), inertia keeps s^2 of a scale error s, and centroids enter $C(\theta)$ itself, corrupting the estimated *masses* by ∼2.4% per millimetre. Were part masses the goal, refinement would be needless by Remark 1; a sim-ready asset also needs density, inertia, and centroids (verified in Sec. IV-B).

C. Information-Theoretic Joint Configuration Selection

The classical limit of static gravity sensing [13] becomes, in density coordinates:

Lemma 1 (*Closed-form information; classical fact restated*): For any finite set $\mathcal{G} = \{\hat{g}_k\}_{k=1}^{K}$ of unit gravity directions, the Fisher information about ρ gathered at configuration θ is

$$M(\theta;\mathcal{G}) = g^2\left[\frac{K}{\sigma_f^2}vv^\top + \frac{1}{\sigma_\tau^2}C^\top\Big(KI_3 - \sum_k \hat{g}_k\hat{g}_k^\top\Big)C\right]. \quad (3)$$

Proof: Expand $A^\top R^{-1}A$ using $[\hat{g}]_\times^\top[\hat{g}]_\times = I_3 - \hat{g}\hat{g}^\top$; sum over k. ■

Corollary 1 (*Triad invariance*): If $\mathcal{G}$ is an orthonormal basis then $\sum_k \hat{g}_k\hat{g}_k^\top = I_3$ and

$$M(\theta) = g^2\left[\tfrac{3}{\sigma_f^2}vv^\top + \tfrac{2}{\sigma_\tau^2}C(\theta)^\top C(\theta)\right], \quad (4)$$

which is invariant to rotating the whole triad by any $Q \in SO(3)$.

Cor. 1 collapses the design space to Θ: wrist tilt carries no information, so articulation is the *only* design variable; the freed $SO(3)$ is spent in Sec. III-D. And since $M(\theta)$ is independent of ρ, the loop exists only to absorb angle error.

Proposition 1 (*Rank ceiling*): For any K and any arrangement of gravity directions, $\operatorname{rank} M(\theta;\mathcal{G}) \le 1 + \operatorname{rank} C(\theta) \le 1 + \min(3, P)$.

Its proof—PSD ordering plus rank subadditivity—is standard.

Corollary 2 (*Rigid objects are capped at four*): Holding the configuration fixed, no number of gravity directions or repeated measurements raises the rank above four: a rigid assembly of $P \ge 5$ parts is not identifiable from quasi-static wrench sensing without additional priors—the closure rigid-object methods [1], [3], [6] must buy.

Proposition 2 (*What stays invisible*): With an orthonormal triad, $\mathcal{N}(M(\theta)) = \{u : \sum_i u_i V_i = 0 \text{ and } \sum_i u_i V_i c_i(\theta) = 0\}$: a redistribution of mass preserving both the total mass and the first moment about the sensor origin is unobservable at θ.

Proof: With $M = a\,vv^\top + b\,C^\top C$, $a, b > 0$: $Mu = 0 \iff a(v^\top u)^2 + b\|Cu\|^2 = 0 \iff v^\top u = 0,\ C(\theta)u = 0$. ■

A redistribution hidden at θ_1 becomes visible at θ_2 iff $C(\theta_2)u \neq 0$; only articulation moves C—the method is specific to articulated objects, not merely applicable: a redistribution of $+100/-200/+100$ g preserving total mass and first moment is invisible at every wrist orientation, and folding the joint moves the far centroid and reveals it.

Proposition 3 (*Lower bound on rounds*): With $M_R = \sum_{r=1}^{R} M(\theta_r)$ we have $\operatorname{rank} M_R \le 1 + \min(3R, P)$, so identifying all P densities requires $R \ge \lceil (P-1)/3 \rceil$ distinct configurations.

The proof stacks the $C(\theta_r)$ and applies subadditivity to the Gram matrix. For a serial chain of p links with a friction hinge per joint, $P = 2p - 1$:

$$R \ \ge\ \lceil (2p-2)/3 \rceil, \quad (5)$$

met with equality on both custom objects (Sec. IV-C). In density coordinates, the geometric observability characterization of [17], specialized to quasi-static gravity, is exactly Props. 1 and 2; the configuration count of Prop. 3 is what the density view adds.

Remark 2 (*Channel separation*): The force block $-g(v^\top\rho)\hat{g}$ of (2) is free of θ: joint-angle error enters $R_{\text{eff}} = R + J\Sigma_\theta J^\top$ through the torque rows alone. Rank one with row space $\operatorname{span}\{v\}$, it determines only the total mass the scale provides; the part-separating and angle-corrupted subspaces coincide, so reducing σ_f improves nothing the scale has not fixed.

Remark 3 (*Necessary, not sufficient*): Prop. 3 constrains rank only and grows *additively*: two unknowns per link buy at most a third of a round. Once M_R is full rank, conditioning governs, degrading *multiplicatively*—roughly twofold per link, hence with half-width $\propto 1/\sqrt{R}$ roughly fourfold more rounds per link (quantified in Sec. IV-C).

Each round maximizes expected information gain under the current posterior Σ,

$$\theta^\star = \arg\max_{\theta\in\Theta_{\text{feas}}} \ \tfrac{1}{2}\log\det\big(I + R_{\text{eff}}^{-1/2}A(\theta)\Sigma A(\theta)^\top R_{\text{eff}}^{-1/2}\big), \quad (6)$$

on the relative-uncertainty scaling of Σ. The triad drops out (Cor. 1): the search runs over joint angles only, by multi-start optimization over Θ_{feas} of Sec. III-D. (6) is classical D-optimality [18], [19]; ours differs in procedure, not criterion: the design variable is the object's own articulation, not the robot's trajectory, and Σ is the per-round posterior, not a prior optimized once.

The accepted $\theta^\star$ goes to the operator: the robot holds still, the interface shows the target angles, and the operator sets the joints by hand—the one intervention PIVoT adds beyond reconstruction; the robot proceeds only after the camera reads the achieved angles.

D. Multi-Topology Measurement Path Design

Each configuration is measured at three poses forming an orthonormal gravity triad—not for observability: two non-parallel directions already attain the rank ceiling of Prop. 1 (for $K = 2$ the weighting in (3) is positive definite). The third direction completes $\sum_k \hat{g}_k\hat{g}_k^\top = I_3$ (Cor. 1): information becomes isotropic, the ceiling is met at the direction set's best conditioning, and the triad's $SO(3)$ orientation becomes informationally *free*. Measurement agrees (Sec. IV-D).

The free orientation pays for manipulation: rotating the whole triad changes nothing in (4), so we rotate it for grasp feasibility, reachability, collision avoidance, and joint protection at zero informational cost (quantified in Sec. IV-D).

Because θ changes the object's *shape*, feasibility is rechecked per candidate—identification design under restricted trajectories, with a shape-changing payload: $\theta^\star$ must admit grasping and holding at all three poses without collision, across the tracker's angular-uncertainty interval, plus a collision-free inter-pose path *under the shape at that configuration* (planning against the previous round's shape collides in transit). Failing candidates are excluded with a neighbourhood; (6) is re-solved.

Paths are planned by RRT-Connect in the arm's joint space with the object rigidly attached, checking joint limits, environment and self-collisions, and the object's own collision pairs. Topology changes the problem, not its parameters—spatial extent, active self-collision pairs, reachable reorientations—so triad rotation and paths are re-solved per topology and configuration. Joint visibility, too: an axis near the camera's line of sight projects poorly (worst 0.37 px/deg, a 4.1× spread), so each joint is read where its axis is best conditioned.

E. Active Sensing and Uncertainty-Aware Density Update

The robot keeps inertial forces below one percent of gravity, holds still at each pose, and reads the wrench after tare. The regressed configuration is the camera-*measured* $\hat\theta$, so $A(\hat\theta)$ is corrupted: errors-in-variables, under which weighted least squares stays biased however many rounds accumulate.

We therefore estimate the per-round angle correction δ_r jointly with ρ:

$$\min_{\rho,\{\delta_r\}} \sum_r \left\| y_r - A(\hat\theta_r+\delta_r)\rho \right\|^2_{R_{\text{eff}}^{-1}} + \sum_r \|\delta_r\|^2_{\Sigma_\theta^{-1}} + \|\rho-\mu_0\|^2_{\Sigma_0^{-1}}, \tag{7}$$

under positivity box constraints, with Σ_θ the tracker's angle covariance and the prior seeded by the scale—the total-least-squares remedy applied to the design variable itself. A grasp offset δ adds $MG[\hat g]_\times\delta$ to the torque rows—three linear unknowns, M known from the scale. The posterior propagates with $R_{\text{eff}} = R + J\Sigma_\theta J^\top$, $J = \partial(A\rho)/\partial\theta$.

Because (7) fits $y \approx A(\hat\theta+\delta)\rho$, residuals must be scored at $\hat\theta_r + \delta_r$: scoring at $\hat\theta$ charges the recovered correction back to the residual—interval inflation of 79× at $p = 4$, the stopping test never satisfied though the estimate is accurate to 0.28%—a structural trap of pairing an errors-in-variables estimator with a residual-based stopping rule.

The loop stops when the 95% relative half-width over the *parts* falls below $\tau(p) = 0.5\% \times p$ (1.0–3.0% for $p = 2$–6), inflated when the residual exceeds the noise prediction; the per-part budget follows from Remark 3. Hinge rows known from the scale are excluded lest their prior width block termination, and the iteration budget scales with $P + RJ$. Unmet, Σ re-enters (6).

One quantity is beyond this loop by physics, not design: the inertia tensor cannot be identified quasi-statically—at rest $\omega = 0$ and $\alpha = 0$, so every term of $\tau = I\alpha + \omega\times(I\omega)$ containing I vanishes identically, and the static wrench constrains only the zeroth and first mass moments (Prop. 1). Exported inertia is *derived*—mesh geometry times identified density under the uniform-density assumption—not measured, inheriting volume-scale error as s^2 (Sec. III-B) and bounding what any quasi-static method can claim.

The converged $\hat\rho$ yields each part's mass, center of mass, and derived inertia analytically, written into the URDF with geometry, joints, and Gaussians untouched; the asset is revalidated in simulation at measured and held-out configurations.

TABLE I
EVALUATION OBJECTS: LINKS, INDEPENDENTLY UNKNOWN DENSITIES P, SINGLE-CONFIGURATION RANK $1 + \min(3, P)$ (PROP. 1), CONFIGURATION LOWER BOUND $R_{\min} = \lceil (P-1)/3 \rceil$ (PROP. 3), AND GROUND-TRUTH SOURCE.

Object	Links	P	rank	$R_{\min}$	Ground truth
Custom 2-link	2	3[h]	3	1	CAD + material
Custom 3-link	3	5[h]	4	2	CAD + material
Stand lamp	3	3	3	1	scan + disassembly
Laptop[†]	2	2	3	1	scale + part split

Rank and $R_{\min}$ follow from geometry alone, before any wrench is measured. [h] *includes the friction hinge as its own unknown (41 g each). The hinge occupies 7.4 cm^3 at 5.51 g/cm^3—small, largely occluded, and mass-dense, the case that makes part-wise identification hard rather than the case that makes it easy.* [†] *Laptop mesh and URDF are in preparation; it contributes ground truth and vision-baseline rows only.*

IV. EXPERIMENTS

A. Setup

Fig. 3 Experimental setup and objects [PLACEHOLDER]

RB5-850E + AFT200-D80-C wrist F/T + Robotiq 2F-85 + RealSense D456,
beside the same scene in Drake.
Four objects with parts colour-coded; hinge parts marked.

Fig. 3. Experimental platform and the four evaluation objects of Table I. [Placeholder: the photo awaits the hardware session.]

The platform: a six-axis RB5-850E arm, wrist force/torque sensor, Robotiq 2F-85 gripper, and one RealSense D456; joint angles are tracked on the RGB-D stream—measured, not commanded (Fig. 3). All results below are Drake simulation, running the estimator, selection rule, and stopping test of Sec. III-E unchanged; the hardware runs are in preparation, and every real-robot quantity is marked pending.

We evaluate four articulated objects (Fig. 3, Table I): two custom 3D prints with per-part densities we set, a disassembly-weighed stand lamp from a 3DGS scan, and a laptop; the custom objects give their friction hinge its own part—the harder general case. Density identification covers three of the four: the laptop's mesh and URDF assets are in preparation, so it contributes ground-truth and vision-baseline rows only.

Metrics: per-part relative density error; 95% half-width and its coverage (a confident wrong answer is worse than an uncertain one); rounds to the target half-width $\tau = 0.5\% \times P$; asset trajectory RMSE. All methods share the same fixed part volumes, so density error and mass error are the same number (Table II footnotes). Four objects admit no statistical-significance claim, so tables show per-part rows, not pooled means.

Baselines are PUGS [3], SiPhy [11], and PhysX-Omni [5] from vision, plus three ablations: *uniform* (one density per assembly [1]), *single* (one fixed configuration, unlimited re-orientations and repeats [6]), and *WLS* (our loop, ignoring the regressor's own angle error, Eq. (7)). Vision outputs lacking per-part densities are converted in whichever direction favors the baseline (Table II footnotes).

B. Part-Wise Volume Estimation

The CAD solids and displacement measurements are scoring references, never inputs. The mesh refinement and per-part scaling implementation is being delivered by a collaborator, so the measured per-part volume-error table is deferred; no claim below depends on it. What we can already quantify is the downstream cost: we perturb the volumes the estimator *believes* by $\pm\delta$ with random signs while measurements come from the true plant, scoring masses with the volumes actually used ($\delta \leq 30\%$; Fig. 4).

Density error follows volume error near-exactly one-to-one; part mass is far less sensitive but not exactly invariant —3–6% at $\delta = 30\%$, the MAP prior's cancellation being inexact (Remark 1 covers only the least-squares core)— and the mesh-derived inertia tensor degrades as s^2 (19.1% at $\delta = 30\%$).

Volume error also costs *rounds*: the worst part drags the loop, median rounds growing from 1 to 7 on the 2-link object by $\delta = 30\%$, with three of eight 3-link seeds failing to converge there. Refinement is what keeps the interaction count bounded.

C. Part-Wise Density Identification

Each object runs the closed loop over eight seeds (5% joint-angle error, $\tau = 0.5\% \times P$); Table II is the main result. Prop. 3 reads off geometry (Table I): the 3-link object's five unknowns exceed the rank-four ceiling, so a second configuration is *necessary* before any wrench is simulated.

The loop meets that bound (one round on the 2-link object, two elsewhere; all seeds converge). Uniform fails by orders of magnitude: the mass redistributions it misses stay invisible to shape until a second articulation configuration (Prop. 2).

On the 2-link object *ours is single*: three unknowns fit under the rank-four ceiling, and the two columns of Table II are byte-identical—the positive direction of Prop. 3, not a win; only uniform is separated there. Single's failure appears on the 3-link object, in the half-width rather than the point error: its worst-part error is a respectable 1.58%, but its half-widths for the three links are 2.6, 28.7, and 42.5%—one configuration cannot *certify* five unknowns.

The strongest ablation evidence is the coverage block of Table II. WLS shares our loop, ignoring only the errors-in-variables structure; its point errors stay within an order of magnitude of ours, but its 95% intervals miss the truth on up to 20 of 24 part–seed pairs where ours cover at target width: the bias Eq. (7) removes does not shrink with rounds. Single's coverage comes only from width. An estimator feeding a simulator must know when it is wrong.

The hinge rows are the built-for case: densities recovered to 0.04–0.34%, intervals covering on all seeds, at 3.0–3.8% half-widths honestly reflecting the occluded part's weaker observability. The stand lamp has no hinge body, so this rests on the custom objects only.

The stand lamp attains full rank in one configuration yet runs two: past full rank, conditioning governs, set by joint-angle precision rather than rank (Remark 3, Fig. 5).

Synthetic chains of $P{=}2\dots6$ unknowns separate the methods cleanly. Through $P = 3$ everything but uniform suffices; from $P = 4$ the rank-four ceiling of Cor. 2 appears directly in *single*, whose worst-part error climbs $17.06 \rightarrow 32.02 \rightarrow 53.57\%$ as P grows past four, while WLS degrades $10.53 \rightarrow 11.79 \rightarrow 21.39\%$ and ours stays below 0.6% throughout. At $P = 4$ ours converges in a median of 3 rounds where offline D-optimal design [18], [19] needs 4 and random selection needs 21 and fails once in eight; at $P = 5$ the gap widens to 7 against 10 and random never converges (0/8). All three share the criterion, so closed-loop re-selection is the only difference. The $P = 6$ arm is still computing. Beyond $P = 5$, though, no criterion—D, A, or E—reaches a 1% target in twelve rounds: at ±5% angular error the binding resource is angular precision, not the criterion.

D. Measurement Pose and Path Design

How many gravity directions per round: two orthogonal directions reach the rank ceiling, and the orthonormal triad is the knee of diminishing returns—12.88 nat for one direction, 25.41 for two, 26.35 for the triad, only 27.73 for six—while one direction is anisotropic (2.72×): a gravity-aligned joint axis hides distal mass.

The triad's free $SO(3)$ orientation (Cor. 1) is spent on the joints: rotating it to minimize peak torque on those joints preserves the information *exactly*—26.3535 nat before and after, a 0.0% loss—while cutting peak joint torque from 1.031 to $0.620\,\mathrm{N \cdot m}$, a 39.9% reduction: the quantitative basis for claiming measurement does not endanger the object's joints.

We grid each joint range at five values, plan every pose pair (triad cycles, home transits), and score straight-line interpolation against RRT-Connect (Table III, Fig. 6); RRT-Connect never collides, at negligible path-length and planning-time cost.

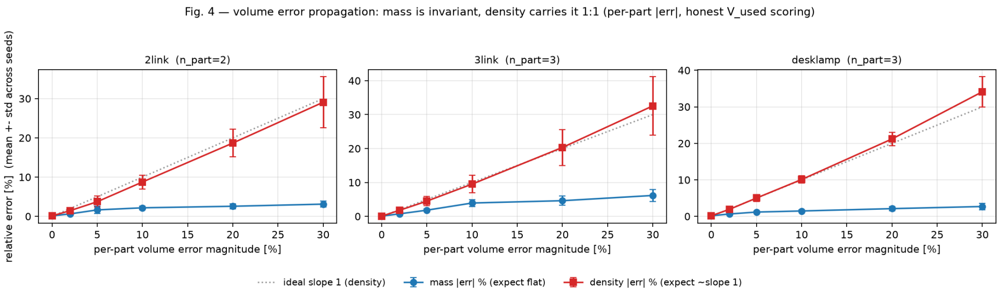

Fig. 4. Injected part-volume error (eight seeds): density error tracks it near one-to-one; part mass, scored with the volumes the estimator used, drifts to 3–6% at $\delta = 30\%$—first-order invariant, not exactly so.

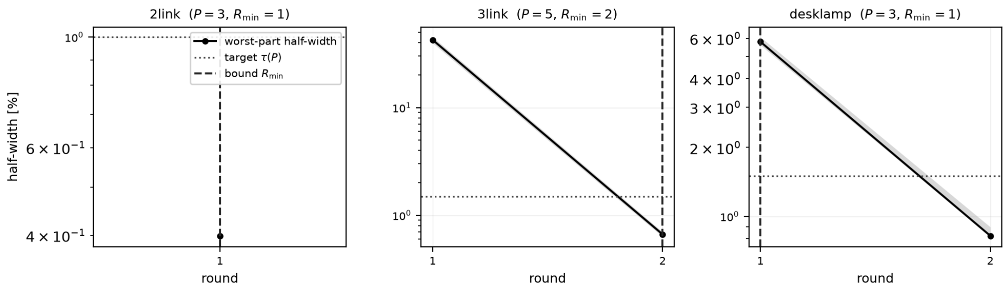

Fig. 5. Worst-part reported half-width per round (median and interquartile range, eight seeds). Dashed: round bound $R_{\min}$ of Prop. 3; dotted: target $\tau(P)$.

TABLE II

PART-WISE DENSITY IDENTIFICATION (SIMULATION). GROUND-TRUTH V, M, $\rho = M/V$; METHOD COLUMNS GIVE THE RELATIVE PER-PART DENSITY ERROR [%] (MEDIAN OVER EIGHT SEEDS FOR INTERACTION METHODS), LOWEST PER ROW IN BOLD.

		Ground truth			Vision baselines			Interaction (median over 8 seeds)			
Object	Part	V [cm^3]	M [g]	ρ [g/cm^3]	PUGS[S]	SiPhy	PhysX-Omni	Uniform	Single	WLS	Ours
Stand lamp	Base	273.00	399.00	1.462	164.1	38.3	69.1	7.33	0.12	0.12	**0.01**
	Support	84.50	85.00	1.006	10.7	27.3	183.1	34.64	1.49	1.27	**0.23**
	Head	64.10	87.00	1.357	36.4	48.5	58.0	0.21	0.73	0.55	**0.08**
2-link*	Long	135.72	49.00	0.361	7887.4	593.4	–	49.14	**0.18**	0.21	**0.18**
	Short	108.12	42.00	0.389	7323.6	544.5	–	25.43	**0.06**	0.10	**0.06**
	Hinge[h]	7.45	41.00	5.506	423.8	54.5	–	51.22	**0.34**	0.35	**0.34**
3-link*	Link 0	288.12	100.00	0.347	4312.5	349.6	–	43.32	0.11	**0.01**	**0.01**
	Link 1	208.40	92.00	0.442	3369.1	253.5	–	110.52	1.14	0.31	**0.06**
	Link 2	162.28	69.00	0.425	3501.8	267.0	–	321.04	1.58	0.48	**0.10**
	Hinge 1[h]	7.45	41.00	5.506	178.2‡	71.7‡	–	46.45	0.08	**0.04**	**0.04**
	Hinge 2[h]	7.45	41.00	5.506			–	46.45	0.08	**0.05**	0.06
Coverage of the reported 95% interval											
Stand lamp					–	–	–	–	24/24	4/24	22/24
2-link					–	–	–	–	16/16	14/16	16/16
3-link					–	–	–	–	24/24	15/24	24/24

All methods share the same fixed part volumes, so density error and mass error are the same number (maximum observed difference 1.7×10^{-16}) and a single column reports both. The bottom block counts, over all part–seed pairs, how often each method's reported 95% interval covers the truth. [h] *hinge modelled as its own unknown.* [S] *Stand-lamp PUGS masses integrate its frozen SAM-partitioned Gaussian volume and point-wise density, scored at the midpoint of its predicted mass range; other PUGS rows and all SiPhy rows distribute each object-level prediction using ground-truth volume fractions (Sec. IV-A). PhysX-Omni is N/A where its generated shape admits no canonical part mapping.* ‡ *vision baselines were scored on the two hinges jointly.* * *preliminary pre-ballast custom-object masses and densities; CAD volumes are fixed. On the 2-link object Ours and Single are byte-identical (Sec. IV-C). Single's intervals cover only because they are wide (Sec. IV-C), and its convergence flag never consults the stopping rule. The fourth object, the laptop (display / base: $V = 354.10/798.60\,\text{cm}^3$, $M = 379.25/1137.75\,\text{g}$, $\rho = 1.071/1.425\,\text{g/cm}^3$; vision errors PUGS 76.7/30.1, SiPhy 182.5/108.0, PhysX-Omni 417.3/611.8 %), contributes ground-truth and vision-baseline data only: its mesh and URDF are in preparation, so all its interaction columns are pending.*

The worst *table* penetration is 33.1 mm (3-link tip link), while the larger 93.8 and 83.5 mm figures are the object penetrating the robot's own wrist and forearm. Joint-limit violations are zero for *both* planners, necessarily (the limits form a convex box): interpolation fails by collision, never by limit violation. Nor always—the 3-link transits are straight-line collision-free—so feasibility is re-checked per configuration.

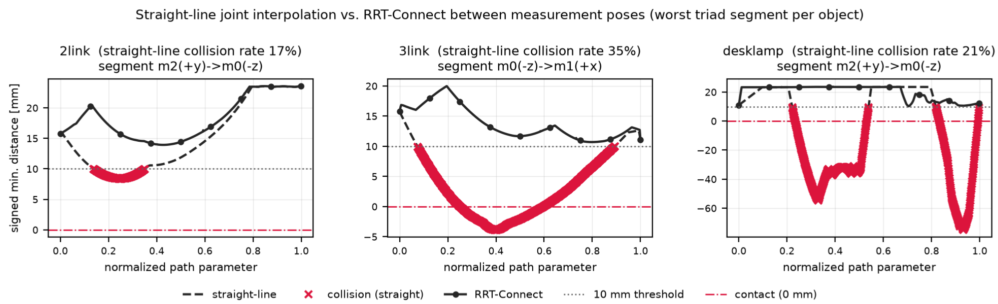

Fig. 6. One row per object at its most informative colliding configuration: joint-space paths, minimum clearance (dashed: straight; solid: RRT-Connect), and end-link workspace trace.

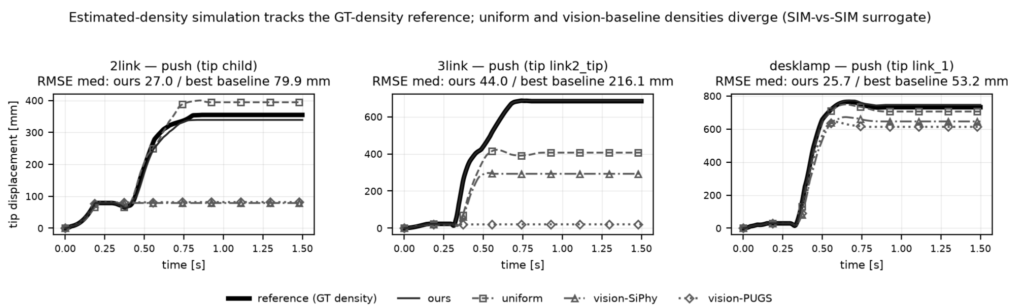

Fig. 7. Asset dynamic fidelity (*simulation-only surrogate*): tip displacement in three scenarios against the ground-truth-density reference plant. [Sim-vs-real version awaits the hardware sessions.]

TABLE III
STRAIGHT-LINE JOINT INTERPOLATION VS. RRT-CONNECT OVER EVERY POSE PAIR OF THE CONFIGURATION GRID (EACH CELL: STRAIGHT / RRT-CONNECT), WITH PER-CONFIGURATION INFORMATION GAIN ΔI AND WHITENED-REGRESSOR CONDITION NUMBER κ.

Object	ΔI [nat] / cond. κ	Coll. [%]	Pen. [mm]	Clear. [mm]	Time [s]
2-link	4.0–4.5 57–324	25.0 / **0.0**	83.5 / **0.0**	0.6 / 13.1	∼0 / 0.12
3-link	6.3–11.1 10^{16}–10^{18}	20.9 / **0.0**	33.1 / **0.0**	11.3 / 14.1	∼0 / 0.11
Stand lamp	6.9–9.3 8.9–100	21.1 / **0.0**	93.8 / **0.0**	−0.8 / 12.0	∼0 / 0.43

Coll. = fraction of pose pairs whose path touches an obstacle; Pen. = maximum penetration depth; Clear. = mean minimum clearance (negative = inside an obstacle on average); Time = mean planning time per pair. Both planners connect the same measurement poses, so the wrench data and hence the density estimate (Table II) are unchanged; what differs is whether the path is executable without contact. Joint-limit violations are zero for both planners by construction (Sec. IV-D). The 3-link condition numbers are structural, not numerical: one configuration observes total mass and first moment (rank 4) against five unknowns, the rank-four ceiling of Prop. 1 appearing directly in the conditioning; the two three-unknown objects are full rank.

TABLE IV
SIM-READY ASSET VALIDATION, A *SIMULATION-ONLY SURROGATE*: MEDIAN WORST-PART TRAJECTORY RMSE [MM] AGAINST THE GROUND-TRUTH-DENSITY REFERENCE PLANT, PUSH SCENARIO OF FIG. 7, WITH THE USER INTERVENTION EACH ASSET REQUIRES.

	Worst-part RMSE [mm], PUSH vs. GT-density plant (simulation)			
Object	Ours	Uniform	SiPhy	PUGS
2-link	**26.97**	79.94	338.68	342.36
3-link	**43.97**	216.13	491.38	582.44
Stand lamp	**25.66**	53.16	325.22	363.97
User intervention				Time
Ours	as RORA + scale + joint adj. (1–2×)			–
Uniform	as RORA + scale; no interaction			–
SiPhy	single RGB image; no interaction			–
PUGS	multi-view scan; no interaction			–

The reference is a plant carrying the ground-truth densities, not *the real object; contact stiffness, dissipation, friction, and joint damping are shared constants across all variants, so only the per-part densities differ. Medians are over five pre-registered initial conditions; release and drop are summarized in the text (Sec. IV-E). End-to-end wall-clock time requires the hardware runs and is pending. Vision-baseline densities carry the relative errors of Table II with an assumed positive sign; PhysX-Omni is excluded because it is N/A on both custom objects.*

E. Sim-Ready Asset Validation

This validation is a *simulation-only surrogate*: the reference is a Drake plant carrying the ground-truth densities, not the real object, and nothing here is a real-world claim. The sim-to-real comparison (Fig. 7) requires the hardware sessions and is pending. Contact, friction, and damping parameters are shared constants, so the comparison isolates exactly what we identify: the per-part densities.

Each variant runs release, drop, and push; ours ranks first in all 27 scenario cells. Table IV reports the push rows matching Fig. 7; release and drop preserve the same ordering at larger margins—orders of magnitude, versus 2.1–14.2× on push, where contact amplifies every variant's error, ours included: a ranking, not absolute accuracy.

The 2-link release must not be over-read: it is structurally blind to density—with the root link welded, the single pendulum's motion is invariant to scaling the moving link's density, so its sub-millimetre RMSE reflects that invariance, not skill—and release trajectories settle onto joint limits, under-stating differences.

Intervention is part of the asset's specification (Table IV): the addition over the inherited RORA [2] stage is the joint adjustment, once per round—one to two rounds here (operator and wall-clock times await the hardware sessions).

V. Conclusion

A sim-ready articulated asset requires per-part volume *and* density; vision-based estimation is inaccurate, and interaction-based identification rigid-only. PIVoT identifies per-part density in a closed loop: information-gain joint-configuration selection, three gravity-direction measurement poses linked by collision-free paths, and uncertainty-aware updates.

On three simulated objects, worst-part density error was 0.18, 0.11, and 0.23%, versus a homogeneous baseline's 49.1, 321.0, and 34.6%. Stronger evidence is the interval: our 95% intervals covered the truth in 62 of 64 part–seed pairs, a WLS ablation in only 33 of 64—the errors-in-variables bias does not shrink with rounds, and a WLS interval cannot see it—and a single configuration cannot certify five unknowns, its half-widths reaching 42.5% (Cor. 2). Omitting the custom objects' friction hinge as its own unknown yields a 79.1% hinge error under a claimed 0.83% half-width. The intervention added for physics estimation is one joint-configuration adjustment; the reconstruction-stage intervention is inherited from [2]. Validation on the laptop asset and the physical robot is in preparation.

Limitations begin with adopted components: the input reconstruction's segmentation and joint errors propagate downstream, with its human-in-the-loop intervention [2]; sampling-based RRT grows costly on complex topologies without optimality guarantees; and a single linkage gripper on a 6-DoF arm restricts validation to small, light objects. The experiments add two. Volume error is not benign: density error tracks it one-to-one, inertia as s^2 (Rem. 1), part mass is far less sensitive yet not exactly invariant, and it multiplies rounds (Sec. IV-B)—the practical limit is conditioning, set by volume and joint-angle precision, not rank. The inertia tensor is not identified: the quasi-static wrench never sees it, so we derive it from mesh geometry and identified density, inheriting mesh scale error quadratically. Future work: estimate joint-angle dynamics and reorient accordingly, where tools like RPNA [17] become discriminative.

Acknowledgment

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